Analytical Chemistry Cumulative Exam
December 6, 2002
(Statistical and Error Analysis of Data)

1. (20 points) Define precision and define accuracy. What kinds of error can lead to imprecise data and what kinds of error can lead to inaccurate results in an analysis? Explain.

2. Iron for biosynthesis is transported through the bloodstream by the protein transferrin. The iron content of transferrin in blood was determined. Four analyses were carried out on a sample using a newly designed method of analysis and yielded the following results (in μg of Fe per mL): **1.05, 0.997, 1.11, 0.800**.

(a) (20 points) Is it appropriate at the 95% confidence level to reject the 0.800 as a gross error?
(From this point onward, if the 0.800 can be rejected, do so and do not retain it in your calculations. If it is to be retained, keep it in all of your calculations.)

(b) (20 points) The accepted value of iron for the sample is **1.15**. Is there bias in this new method of analysis at 95%?

(c) (20 points) A new treatment for people with low blood iron is to be tested on the same person who provided the blood sample analyzed above. A new sample of blood is taken after the treatment and is analyzed. The new results are:
1.22, 1.18, 1.15. Did the treatment have a significant effect on the iron content in blood at the 95% confidence level?

(d) (20 points) After one *hundred* of subsequent analyses with the new method, it is found that the σ value is **0.05** μg of Fe per mL. The old method of iron analysis gave a σ value of **0.06** μg of Fe per mL. Is the new method significantly more precise than the old method at 95% confidence?

Equations and Data that might or might not be helpful.

$$\mu = \bar{x} \pm \frac{t\,s}{\sqrt{N}}$$

$$Q = \frac{\text{gap}}{\text{range}}$$

$$F = \frac{s_1^2}{s_2^2}$$

$$s = \sqrt{\frac{\sum\left(x_i - \bar{x}\right)^2}{N-1}}$$

$$\sigma = \sqrt{\frac{\sum\left(x_i - \mu\right)^2}{N}}$$

$$\pm t = \left[\left(\bar{x}_a - \bar{x}_b\right) / S_p\right]\sqrt{\frac{N_a N_b}{N_a + N_b}}$$

$$s_p = \sqrt{\frac{\sum\left(x_{ai} + \bar{x}_a\right)^2 + \sum\left(x_{bi} + \bar{x}_b\right)^2}{N_a + N_b - 2}}$$

C. STUDENT t-TEST TABLE

	t-FACTOR FOR CONFIDENCE INTERVAL					
DF	CL = 80% α=0.20	CL = 90% α=0.10	CL = 95% α=0.05	CL = 99% α=0.01	CL = 99.9% α=0.001	
1	3.08	6.31	12.7	63.7	637	
2	1.89	2.92	4.3	9.92	31.6	
3	1.64	2.35	3.18	5.84	12.9	
4	1.53	2.13	2.78	4.60	8.60	
5	1.48	2.02	2.57	4.03	6.86	
6	1.44	1.94	2.45	3.71	5.96	
7	1.42	1.90	2.36	3.50	5.40	
8	1.40	1.86	2.31	3.36	5.04	
9	1.38	1.83	2.26	3.25	4.78	
10	1.37	1.81	2.23	3.17	4.59	
11	1.36	1.80	2.30	3.11	4.44	
12	1.36	1.78	2.18	3.06	4.32	
13	1.35	1.77	2.16	3.01	4.22	
14	1.34	1.76	2.14	2.98	4.14	
∞	1.29	1.64	1.96	2.58	3.29	

Q values

	Level of confidence		
Number of Measurements	90%	95%	99%
3	0.941	0.970	0.994
4	0.765	0.829	0.926
5	0.642	0.710	0.821
6	0.560	0.625	0.740
7	0.507	0.568	0.680
8	0.468	0.526	0.634
9	0.437	0.493	0.598
10	0.412	0.466	0.568
15	0.338	0.384	0.475
20	0.300	0.342	0.425
30	0.260	0.298	0.372

F-Table at 95% Confidence

Degrees of Freedom in Denominator	Degrees of Freedom in Numerator										
	1	2	3	4	5	6	7	8	9	10	100
1.	161.448	18.513	10.128	7.709	6.608	5.987	5.591	5.318	5.117	4.965	
2.	199.500	19.000	9.552	6.944	5.786	5.143	4.737	4.459	4.256	4.103	
3.	215.707	19.164	9.277	6.591	5.409	4.757	4.347	4.066	3.863	3.708	8.55
4.	224.583	19.247	9.117	6.388	5.192	4.534	4.120	3.838	3.633	3.478	5.66
5.	230.162	19.296	9.013	6.256	5.050	4.387	3.972	3.687	3.482	3.326	4.41
6.	233.986	19.330	8.941	6.163	4.950	4.284	3.866	3.581	3.374	3.217	3.71
7.	236.768	19.353	8.887	6.094	4.876	4.207	3.787	3.500	3.293	3.135	3.27
8.	238.882	19.371	8.845	6.041	4.818	4.147	3.726	3.438	3.230	3.072	2.97
9.	240.543	19.385	8.812	5.999	4.772	4.099	3.677	3.388	3.179	3.020	2.76
10.	41.882	19.396	8.786	5.964	4.735	4.060	3.637	3.347	3.137	2.978	2.59
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93	1.39